

1. LeRobot: What is it, and why is Hugging Face building a unified library for robotics? How does it differ from traditional control stacks?

LeRobot is basically a robotics library developed by Hugging Face. As in other popular deep learning libraries, it is based on PyTorch.

I think we can approach the LeRobot library just like Ultralytics. Think about the impact that Ultralytics made on computer vision. With a few lines, anybody can detect objects, or with approximately 50 lines, anybody can train a custom object detection model. Even though people don't have any knowledge about computer vision, they started to integrate object detection, segmentation, and keypoint detection models into their pipelines thanks to Ultralytics, because there is nothing complex anymore.

I believe Hugging Face is trying to do exactly the same thing. They want more people involved in this area, and for that, they are trying to build a user-friendly environment.

And they already did a similar thing by developing the Transformers library for NLP. You only need to install a few libraries with pip, then import them, and anybody can use pretrained models with just a few lines.

Traditional robotics pipelines rely heavily on classical, model-based control paradigms, but LeRobot is more like a data-driven stack. Instead of using explicit rules, now there are models. I believe, just like in computer vision, the main task is generalization with minimal data. Yes, we can provide more data and obtain better models, but the main focus will be how to do it with the least amount of data and still generalize well.

LeRobot integrates everything into one package: hardware, software, and training. It handles the entire process, and it supports my earlier points.

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2. SmolVLA: Research the general architecture. How does a VLA model differ from a standard LLM or Vision model in a robotics context?

SmolVLA is a Vision-Language-Action model developed by Hugging Face as part of the LeRobot ecosystem. One of the most important things about SmolVLA is that it is lightweight. I had an old GPU (6GB VRAM), and with CUDA support, I managed to run fine-tuned models on the GPU. Unlike other models, you don't need powerful GPUs, and this makes it accessible for most people.

VLMs work like this: you provide an image input and a text input, and they give you a result. The result can be a bounding box, mask, or text (context of the image). Whereas in VLAs, they understand the environment, understand the text prompt, and act. This action part is like the movement of axes and joints to complete a task. It outputs commands for the robot's hardware to perform some movement.

SmolVLA uses a streamlined vision-language model (specifically SmolVLM2, which combines a SigLIP vision encoder and a SmolLM2 language decoder) to process camera

feeds and text instructions. Then it attaches a Flow-Matching Transformer to the VLM, which translates the VLM's understanding into robotic arm motor commands.

The outputs of LLMs and VLMs can be meaningful most of the time, but not those of VLAs. They are more like angles for specific joints, which are hard to interpret, especially when there are many axes in a robotic arm.